Module 4: Key Terms & Course Overview

Terms and Definitions from Course 1, Module 4

- **Antivirus software:** A software program used to prevent, detect, and eliminate malware and viruses [cite: 75].
- **Database:** An organized collection of information or data [cite: 75].
- **Data point:** A specific piece of information [cite: 75].
- **Intrusion detection system (IDS):** An application that monitors system activity and alerts on possible intrusions [cite: 75].
- **Linux:** An open-source operating system [cite: 75].
- **Log:** A record of events that occur within an organization's systems [cite: 75].
- **Network protocol analyzer (packet sniffer):** A tool designed to capture and analyze data traffic within a network [cite: 75].
- **Order of volatility:** A sequence outlining the order of data that must be preserved from first to last [cite: 75].
- **Programming:** A process that can be used to create a specific set of instructions for a computer to execute tasks [cite: 75].
- **Protecting and preserving evidence:** The process of properly working with fragile and volatile digital evidence [cite: 75].
- **Security information and event management (SIEM):** An application that collects and analyzes log data to monitor critical activities in an organization [cite: 75].
- **SQL (Structured Query Language):** A query language used to create, interact with, and request information from a database [cite: 75].

Google Cybersecurity Certificate Course Overview

Congratulations on completing Course 1 of the Google Cybersecurity Certificate: **Foundations of Cybersecurity!** [cite: 76]. In this part of the program, you learned about possible career paths and key skills for cybersecurity professionals, and were introduced to foundational cybersecurity terms and concepts [cite: 76].

The program consists of the following courses [cite: 76, 77]:

1. **Foundations of Cybersecurity** — Explore the cybersecurity profession, including significant events that led to the development of the cybersecurity field and its continued importance to organizational operations. Learn about entry-level cybersecurity roles and responsibilities [cite: 76].
2. **Play It Safe: Manage Security Risks** — Identify how cybersecurity professionals use frameworks and controls to protect business operations, and explore common cybersecurity tools [cite: 76].
3. **Connect and Protect: Networks and Network Security** — Gain an understanding of network-level vulnerabilities and how to secure networks [cite: 76].
4. **Tools of the Trade: Linux and SQL** — Explore foundational computing skills, including communicating with the Linux operating system through the command line and querying databases with SQL [cite: 76].
5. **Assets, Threats, and Vulnerabilities** — Learn about the importance of security controls and developing a threat actor mindset to protect and defend an organization’s assets from various threats, risks, and vulnerabilities [cite: 76].
6. **Sound the Alarm: Detection and Response** — Understand the incident response lifecycle and practice using tools to detect and respond to cybersecurity incidents [cite: 76].
7. **Automate Cybersecurity Tasks with Python** — Explore the Python programming language and write code to automate cybersecurity tasks [cite: 76].
8. **Put It to Work: Prepare for Cybersecurity Jobs** — Learn about incident classification, escalation, and ways to communicate with stakeholders. This course closes out the program with tips on how to engage with the cybersecurity community and prepare for your job search [cite: 77].
9. **Accelerate Your Job Search with AI** — Gain practical job search strategies and learn how to leverage AI tools (like Gemini and Gemini Notebook) to uncover your most valuable skills, create a job search plan, manage your applications, and practice for interviews as you navigate your path to your next role [cite: 77].